JPHG02 Heat and internal energy — Quiz 4

S3 Physics · 3 multiple-choice items

JPHG02.2 Question 1

Which of the following best describes heat?

- **A.** The temperature of an object
- **B.** Energy stored inside an object because of its mass
- **C.** Energy transferred from one body to another because of a temperature difference
- **D.** The average kinetic energy of the molecules of an object

JPHG02.4 Question 2

The same amount of energy is supplied to equal masses of copper and glass. The temperature of the copper rises more than the temperature of the glass. Which statement explains this?

- **A.** Copper has a smaller specific heat capacity than glass.
- **B.** Copper has a larger specific heat capacity than glass.
- **C.** Copper has a higher melting point than glass.
- **D.** Glass does not change temperature when it absorbs energy.

JPHG02.3 Question 3

A 600 W heater heats 0.40 kg of water for 70 s. The specific heat capacity of water is 4 200 J kg^{-1} $°C^{-1}$. All of the electrical energy heats the water. What is the temperature rise?

- **A.** 0.417 °C
- **B.** 10.0 °C
- **C.** 25.0 °C
- **D.** 1 500 °C

JPHG02 Heat and internal energy — Quiz 4 · Answers and solutions

Teacher page · 3 items

JPHG02.2 Question 1

Answer: C. Energy transferred from one body to another because of a temperature difference

Solution: Heat is energy transferred from one body to another because of a temperature difference. Temperature is how hot the object is, and that follows the average kinetic energy of its molecules. Energy stored because of mass is part of the internal energy.

JPHG02.4 Question 2

Answer: A. Copper has a smaller specific heat capacity than glass.

Solution: $E = mc\Delta T$. For the same energy and the same mass, a smaller specific heat capacity gives a larger temperature rise. Copper's temperature rises more, so copper has the smaller specific heat capacity.

JPHG02.3 Question 3

Answer: C. 25.0 °C

Solution: $E = Pt = 600\ \text{W} \times 70\ \text{s} = 42\,000\ \text{J}$. $\Delta T = E / (mc) = 42\,000\ \text{J} / (0.40\ \text{kg} \times 4\,200\ \text{J kg}^{-1}\ \text{°C}^{-1})$ = 25.0 °C. Dividing the time by 60 gives 0.417 °C. Omitting the mass gives 10.0 °C. Treating 70 s as 70 min gives 1 500 °C.